

Lirong Zheng

professor

基本信息

姓名：	Lirong Zheng	所属院校：	皇家理工学院
地区：	Sweden	年龄范围：	50岁以上
海外经历：	有	企业经历：	有

学科领域

- 微电子科学与工程
- 通信工程
- 集成电路设计与集成系统
- 物联网工程
- 柔性电子学

研究领域

- 射频集成电路
- 低功耗电路与能量管理
- 物联网通信与M2M
- 物联网体系结构与协议
- 无线通信与智能天线
- RFID与传感器网络
- 印刷电子学与可拉伸电路
- 柔性器件制备与工艺
- 模拟与射频IC设计
- 超大规模集成电路设计
- +6

个人简介

Lirong Zheng received his Ph.D. in electronic system design from the KTH Royal Institute of Technology in 2001. He has held positions as a Research Fellow, Associate Professor, and Full Professor at KTH. He is the Founding Director of iPack VINN Excellence Center of Sweden, Chair Professor in media electronics at KTH since 2006, and an Ericsson expert since 2007. With over 500 publications and 3 books to his name, his research focuses on electronic circuits and systems for wireless sensing, ambient intelligence, IoT, and smart systems. [Google Scholar](<https://scholar.google.com/citations?user=E2rhFPIAAAAJ&hl=sv>)

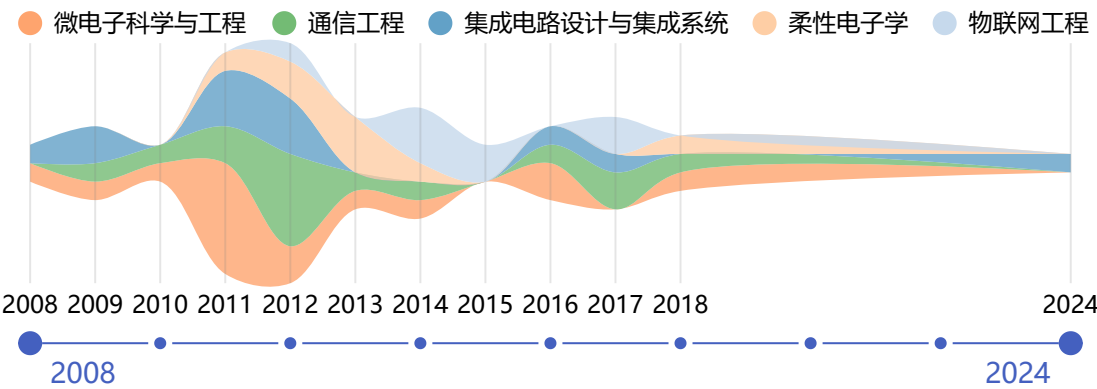
工作经历

Fudan University - Professor and Director	2018-02-01 -
Fudan University - Distinguished Professor, Dean	2010-01-01 -
Royal Institute of Technology - Professor Chair & Director	2006-06-01 -
Royal Institute of Technology - Research Fellow & Docent	1996-11-01 - 2006-06-01
Chinese Academy of Sciences, Shanghai Branch - Assistant Researcher	1992-01-09 - 1996-11-01

教育经历

KTH- Royal Institute of Technology - Ph. D.	1998-02-01 - 2001-10-21
Chinese Academy of Sciences, Shanghai Branch - D. Sc.	1992-09-01 - 1996-05-10
Zhejiang University - B.Sc.	1988-09-01 - 1992-07-01

近年发表的论文



Modeling Cycle-to-Cycle Variation in Memristors for In-Situ Unsupervised Trace-STDP Learning
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS
Xu, Jiawei; Zheng, Yi; Li, Feng; Stathis, Dimitrios; Shen, Ruisi; Chu, Haoming; Lansner, Anders; Zheng, Li-Rong; Zou, Zhuo; Hemani, Ahmed
被引用次数: 0 2024

A Memristor-Based Learning Engine for Synaptic Trace-Based Online Learning
IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS
Wang, Deyu; Xu, Jiawei; Li, Feng; Zhang, Lianhao; Cao, Chengwei; Stathis, Dimitrios; Lansner, Anders; Hemani, Ahmed; Zheng, Li-Rong; Zou, Zhuo
被引用次数: 4 2023

A Design of Autonomous Error-Tolerant Architectures for Massively Parallel Computing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Liu, Lizheng; Jin, Yi; Liu, Yi; Ma, Ning; Huan, Yuxiang; Zou, Zhuo; Zheng, Lirong

被引用次数: 2

2018

Aerosol jet printed silver nanowire transparent electrode for flexible electronic application

JOURNAL OF APPLIED PHYSICS

Tu, Li; Yuan, Sijian; Zhang, Huotian; Wang, Pengfei; Cui, Xiaolei; Wang, Jiao; Zhan, Yi-Qiang; Zheng, Li-Rong

被引用次数: 31

2018

A Wirelessly Powered UWB RFID Sensor Tag With Time-Domain Analog-to-Information Interface

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bao, Dongxuan; Zou, Zhuo; Nejad, Majid Baghaei; Qin, Yajie; Zheng, Li-Rong

被引用次数: 19

2018

A 2.4-GHz ISM RF and UWB hybrid RFID real-time locating system for industrial enterprise Internet of Things

ENTERPRISE INFORMATION SYSTEMS

Zhai, Chuanying; Zou, Zhuo; Zhou, Qin; Mao, Jia; Chen, Qiang; Tenhunen, Hannu; Zheng, Lirong; Xu, Lida

被引用次数: 20

2017

Ammonia gas sensor based on flexible polyaniline films for rapid detection of spoilage in protein-rich foods

JOURNAL OF MATERIALS SCIENCE-MATERIALS IN ELECTRONICS

Matindoust, Samaneh; Farzi, Ali; Nejad, Majid Baghaei; Abadi, Mohammad Hadi Shahrokh; Zou, Zhuo; Zheng, Li-Rong

被引用次数: 77

2017

A Mobile-Based High Sensitivity On-Field Organophosphorus Compounds Detecting System for IoT-Based Food Safety Tracking

JOURNAL OF SENSORS

Jin, Han; Qin, Yajie; Liang, Hao; Wan, Lei; Lan, Hao; Chen, Guoping; Liu, Ran; Zheng, Li-rong; Chiang, Patrick; Hong, Zhi-liang

被引用次数: 3

2017

Interactive UHF/UWB RFID tag for mass customization

INFORMATION SYSTEMS FRONTIERS

Shen, Jue; Majid, Baghaei-Nejad; Xie, Li; Mao, Jia; Pang, Zhibo; Feng, Yi; Xu, Lida; Tenhunen, Hannu; Zou, Zhuo; Zheng, Lirong

被引用次数: 4

2017

A UWB-Based Sensor-to-Time Transmitter for RF-Powered Sensing Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mao, Jia; Zou, Zhuo; Zheng, Li-Rong

被引用次数: 9

2016

Food quality and safety monitoring using gas sensor array in intelligent packaging

SENSOR REVIEW

Matindoust, Samaneh; Baghaei-Nejad, Majid; Abadi, Mohammad Hadi Shahrokh; Zou, Zhuo; Zheng, Li-Rong

被引用次数: 102

2016

A 101.4 GOPS/W Reconfigurable and Scalable Control-Centric Embedded Processor for Domain-Specific Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Huan, Yuxiang; Ma, Ning; Mao, Jia; Blixt, Stefan; Lu, Zhonghai; Zou, Zhuo; Zheng, Li-Rong

被引用次数: 3

2016

Design and implementation of multi-mode routers for large-scale inter-core networks

INTEGRATION-THE VLSI JOURNAL

Ma, Ning; Zou, Zhuo; Lu, Zhonghai; Zheng, Lirong

被引用次数: 3

2016

Design and simulation of a standing wave oscillator based PLL

FRONTIERS OF INFORMATION TECHNOLOGY & ELECTRONIC ENGINEERING

Zhang, Wei; Hu, You-de; Zheng, Li-rong

被引用次数: 4

2016

Design of a terminal solution for integration of in-home health care devices and services towards the Internet-of-Things

ENTERPRISE INFORMATION SYSTEMS

Pang, Zhibo; Zheng, Lirong; Tian, Junzhe; Kao-Walter, Sharon; Dubrova, Elena; Chen, Qiang

被引用次数: 146

2015

Low-Cost Printed Chipless RFID Humidity Sensor Tag for Intelligent Packaging

IEEE SENSORS JOURNAL

Feng, Yi; Xie, Li; Chen, Qiang; Zheng, Li-Rong

被引用次数: 199

2015

Value-centric design of the internet-of-things solution for food supply chain: Value creation, sensor portfolio and information fusion

INFORMATION SYSTEMS FRONTIERS

Pang, Zhibo; Chen, Qiang; Han, Weili; Zheng, Lirong

被引用次数: 186

2015

A Subgigahertz UWB Transmitter With Wireless Clock Harvesting for RF-Powered Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mao, Jia; Zou, Zhuo; Zheng, Li-Rong

被引用次数: 8

2014

Item-Level Indoor Localization With Passive UHF RFID Based on Tag Interaction Analysis

IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS

Zhang, Zhi; Lu, Zhonghai; Saakian, Vardan; Qin, Xing; Chen, Qiang; Zheng, Li-Rong

被引用次数: 83

2014

A Health-IoT Platform Based on the Integration of Intelligent Packaging, Unobtrusive Bio-Sensor, and Intelligent Medicine Box

IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS

Yang, Geng; Xie, Li; Mantysalo, Matti; Zhou, Xiaolin; Pang, Zhibo; Xu, Li Da; Kao-Walter, Sharon; Chen, Qiang; Zheng, Li-Rong

被引用次数: 425

2014

Materials capability and device performance in flexible electronics for the Internet of Things

JOURNAL OF MATERIALS CHEMISTRY C

Zhan, Yiqiang; Mei, Yongfeng; Zheng, Lirong

被引用次数: 155

2014

Radio frequency identification enabled wireless sensing for intelligent food logistics

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Zou, Zhuo; Chen, Qing; Uysal, Ismail; Zheng, Lirong

被引用次数: 55

2014

RFID antenna humidity sensor co-design for USN applications

IEICE ELECTRONICS EXPRESS

Amin, Yasar; Feng, Yi; Chen, Qiang; Zheng, Li-Rong; Tenhunen, Hannu

被引用次数: 5

2013

Electromagnetic Analysis of Radio Frequency Identification Antennas for Green Electronics

ELECTROMAGNETICS

Amin, Yasar; Shao, Botao; Chen, Qiang; Zheng, Li-Rong; Tenhunen, Hannu

被引用次数: 1

2013

A Hybrid Low Power Biopatch for Body Surface Potential Measurement

IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS

Yang, Geng; Chen, Jian; Xie, Li; Mao, Jia; Tenhunen, Hannu; Zheng, Li-Rong

被引用次数: 30

2013

Directly Printed Packaging-Paper-Based Chipless RFID Tag With Coplanar LC Resonator

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Shao, Botao; Amin, Yasar; Chen, Qiang; Liu, Ran; Zheng, Li-Rong

被引用次数: 40

2013

Integration of f-MWCNT Sensor and Printed Circuits on Paper Substrate

IEEE SENSORS JOURNAL

Xie, Li; Feng, Yi; Mantysalo, Matti; Chen, Qiang; Zheng, Li-Rong	
被引用次数: 28	2013
Chipless RFID tags fabricated by fully printing of metallic inks	
ANNALS OF TELECOMMUNICATIONS	
Shao, Botao; Chen, Qiang; Amin, Yasar; Liu, Ran; Zheng, Li-Rong	
被引用次数: 20	2013
Modeling of peak-to-peak core switching noise, output impedance, and decoupling capacitance along a vertical chain of power distribution TSV pairs	
ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING	
Ahmad, Waqar; Chen, Qiang; Zheng, Li-Rong; Tenhunen, Hannu	
被引用次数: 0	2012
Flexible UHF Resistive Humidity Sensors Based on Carbon Nanotubes	
IEEE SENSORS JOURNAL	
Feng, Yi; Cabezas, Ana Lopez; Chen, Qiang; Zheng, Li-Rong; Zhang, Zhi-Bin	
被引用次数: 21	2012
Electrical performance and reliability evaluation of inkjet-printed Ag interconnections on paper substrates	
MATERIALS LETTERS	
Xie, Li; Mantysalo, Matti; Cabezas, Ana Lopez; Feng, Yi; Jonsson, Fredrik; Zheng, Li-Rong	
被引用次数: 38	2012
Green wideband RFID tag antenna for supply chain applications	
IEICE ELECTRONICS EXPRESS	
Amin, Yasar; Kanth, Rajeev Kumar; Liljeberg, Pasi; Chen, Qiang; Zheng, Li-Rong; Tenhunen, Hannu	
被引用次数: 1	2012
Code division multiple access/pulse position modulation ultra-wideband radio frequency identification for Internet of Things: concept and analysis	
INTERNATIONAL JOURNAL OF COMMUNICATION SYSTEMS	
Zhang, Zhi; Lu, Zhonghai; Chen, Qiang; Yan, Xiaolang; Zheng, Li-Rong	
被引用次数: 19	2012
DESIGN OF FULLY PRINTABLE AND CONFIGURABLE CHIPLESS RFID TAG ON FLEXIBLE SUBSTRATE	
MICROWAVE AND OPTICAL TECHNOLOGY LETTERS	
Shao, Botao; Chen, Qiang; Liu, Ran; Zheng, Li-Rong	
被引用次数: 29	2012
Study on Glass-Epoxy-Based Low-Cost and Compact Tip-Truncated Triangular Printed Antenna	
INTERNATIONAL JOURNAL OF ANTENNAS AND PROPAGATION	
Kanth, Rajeev Kumar; Liljeberg, Pasi; Tenhunen, Hannu; Chen, Qiang; Zheng, Lirong; Kumar, Harish	

被引用次数: 3

2012

Performance Analysis of Flow-Based Traffic Splitting Strategy on Cluster-Mesh Sensor Networks

INTERNATIONAL JOURNAL OF DISTRIBUTED SENSOR NETWORKS

She, Huimin; Lu, Zhonghai; Jantsch, Axel; Zhou, Dian; Zheng, Li-Rong

被引用次数: 5

2012

Heterogeneous Integration of Bio-Sensing System-on-Chip and Printed Electronics

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Xie, Li; Yang, Geng; Mantysalo, Matti; Xu, Lin-Lin; Jonsson, Fredrik; Zheng, Li-Rong

被引用次数: 35

2012

The Design of All-Digital Polar Transmitter Based on ADPLL and Phase Synchronized $\Delta\Sigma$ Modulator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chen, Jian; Rong, Liang; Jonsson, Fredrik; Yang, Geng; Zheng, Li-Rong

被引用次数: 33

2012

Peak-to-Peak Ground Noise on a Power Distribution TSV Pair as a Function of Rise Time in 3-D Stack of Dies Interconnected Through TSVs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Ahmad, Waqar; Zheng, Li-Rong; Chen, Qiang; Tenhunen, Hannu

被引用次数: 4

2011

A Low-Power and Flexible Energy Detection IR-UWB Receiver for RFID and Wireless Sensor Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zou, Zhuo; Mendoza, David Sarmiento; Wang, Peng; Zhou, Qin; Mao, Jia; Jonsson, Fredrik; Tenhunen, Hannu; Zheng, Li-Rong

被引用次数: 68

2011

Flicker noise conversion in CMOS LC oscillators: capacitance modulation dominance and core device sizing

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Chen, Jian; Jonsson, Fredrik; Carlsson, Mats; Hedenas, Charlotta; Zheng, Li-Rong

被引用次数: 1

2011

A Low Power, Startup Ensured and Constant Amplitude Class-C VCO in 0.18 μm CMOS

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Chen, Jian; Jonsson, Fredrik; Carlsson, Mats; Hedenas, Charlotta; Zheng, Li-Rong

被引用次数: 47

2011

CONFIGURABLE INK-JET-PRINTED RFID TAG ON PAPER SUBSTRATE FOR LOW COST AND GREEN APPLICATIONS

MICROWAVE AND OPTICAL TECHNOLOGY LETTERS

Shao, Botao; Chen, Qiang; Liu, Ran; Zheng, Li-Rong

被引用次数: 1	2011
On Gate Capacitance of Nanotube Networks	
IEEE ELECTRON DEVICE LETTERS	
Liu, Zhiying; Li, Jiantong; Qiu, Zhi-Jun; Zhang, Zhi-Bin; Zheng, Li-Rong; Zhang, Shi-Li	
被引用次数: 7	2011
Mobility Extraction for Nanotube TFTs	
IEEE ELECTRON DEVICE LETTERS	
Liu, Zhiying; Qiu, Zhi-Jun; Zhang, Zhi-Bin; Zheng, Li-Rong; Zhang, Shi-Li	
被引用次数: 13	2011
Solution-Processable Nanotube/Polymer Composite for High-Performance TFTs	
IEEE ELECTRON DEVICE LETTERS	
Liu, Zhiying; Zhang, Zhi-Bin; Chen, Qiang; Zheng, Li-Rong; Zhang, Shi-Li	
被引用次数: 9	2011
Ink-jet printed thin-film transistors with carbon nanotube channels shaped in long strips	
JOURNAL OF APPLIED PHYSICS	
Li, Jiantong; Unander, Tomas; Cabezas, Ana Lopez; Shao, Botao; Liu, Zhiying; Feng, Yi; Forsberg, Esteban Bernales; Zhang, Zhi-Bin; Jogi, Indrek; Gao, Xindong; Boman, Mats; Zheng, Li-Rong; Ostling, Mikael; Nilsson, Hans-Erik; Zhang, Shi-Li	
被引用次数: 21	2011
Far-Field On-Chip Antennas Monolithically Integrated in a Wireless-Powered 5.8-GHz Downlink/UWB Uplink RFID Tag in 0.18-μm Standard CMOS	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Radiom, Soheil; Baghaei-Nejad, Majid; Mohammadpour-Aghdam, Karim; Vandenbosch, Guy A. E.; Zheng, Li-Rong; Gielen, Georges G. E.	
被引用次数: 68	2010
ARCHER: an automated RF-IC Rx front-end circuit design tool	
ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING	
Rodriguez, Saul; Atallah, Jad G.; Rusu, Ana; Zheng, Li-Rong; Ismail, Mohammed	
被引用次数: 3	2009
Two-Dimensional and Three-Dimensional Integration of Heterogeneous Electronic Systems Under Cost, Performance, and Technological Constraints	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Weerasekera, Roshan; Pamunuwa, Dinesh; Zheng, Li-Rong; Tenhunen, Hannu	
被引用次数: 27	2009
Minimal-power, delay-balanced SMART repeaters for global interconnects in the nanometer regime	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Weerasekera, Roshan; Pamunuwa, Dinesh; Zheng, Li-Rong; Tenhunen, Hannu	

Modeling of on-chip bus switching current and its impact on noise in power supply grid

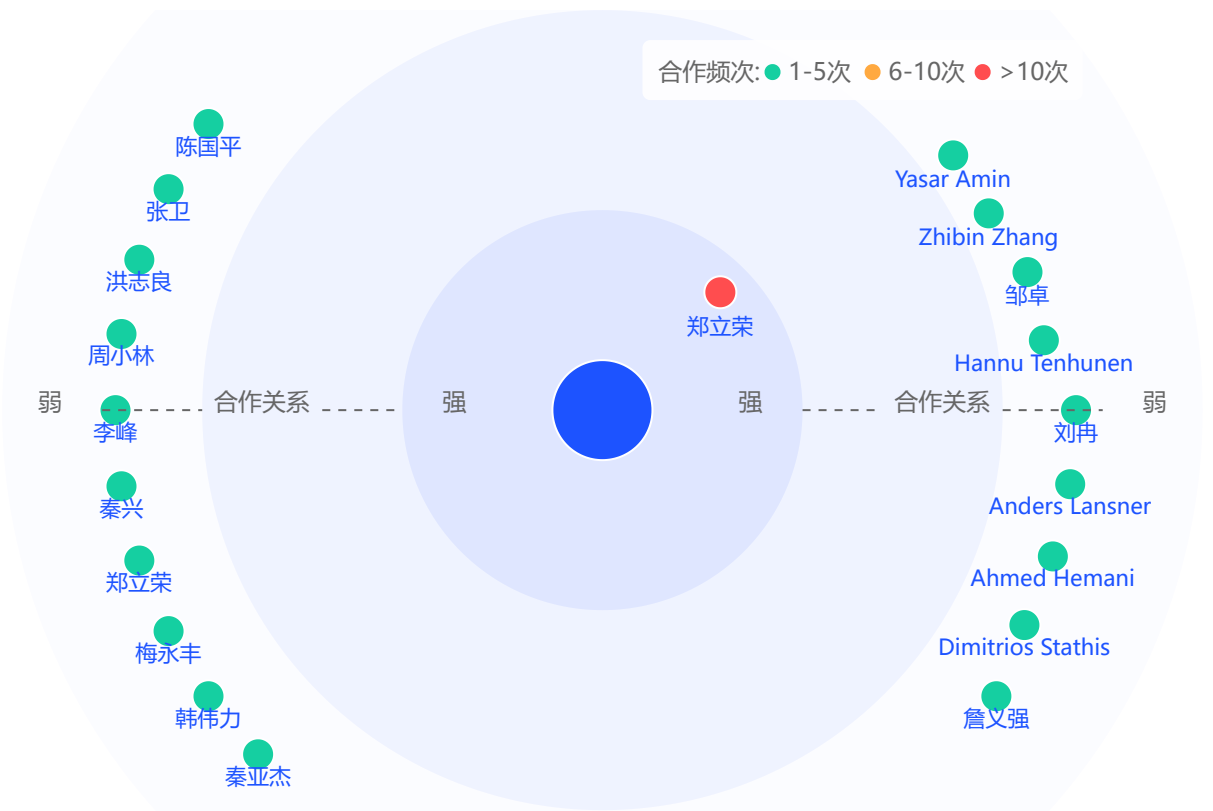
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tuuna, Sampo; Zheng, Li-Rong; Isoaho, Jouni; Tenhunen, Hannu

被引用次数：6

2008

学术合作



郑立荣	合作次数：25
复旦大学	
Yasar Amin	合作次数：5
皇家理工学院	
Zhibin Zhang	合作次数：5
乌普萨拉大学	
邹卓	合作次数：4
复旦大学	
Hannu Tenhunen	合作次数：4
皇家理工学院	

还有 25 个合作者